

Non-Uniqueness of Distribution-Shift Attribution, and Why We Localize Subsets Instead

Technical note

Abstract

For a marketplace comparing an existing batch of data against a new batch, we show a basic *impossibility*: the *total* distribution shift is identifiable and order-invariant, but the *per-feature contribution* to it is not uniquely defined — it depends on the order in which features are considered. We demonstrate this empirically with random feature-addition paths under two metrics: Maximum Mean Discrepancy (MMD) for covariate shift $P(X)$ and the pseudo-outcome / R -risk for concept drift $P(Y | X)$. The consequence is methodological: at any level of the entity hierarchy one should *localize a responsible subset* (well-posed) rather than attribute per-feature proportions (ill-posed). We give the construction, the empirical result, the resulting multi-level drill-down, a worked business example, and how such a system should be evaluated.

1 Setup

Orders (observations) carry raw attributes and are labelled by batch $W \in \{0, 1\}$ (existing vs. new). We roll orders up to an entity axis $\mathbf{e}$ (merchant, buyer, item, region, ...) with a feature map Φ producing an entity–feature matrix $x \in \mathbb{R}^p$ (rich + rolling aggregations), with features grouped by raw attribute, $g : \{1, \dots, p\} \rightarrow \mathcal{G}$. Let P_0, P_1 be the two-batch feature distributions. A characteristic-kernel MMD measures covariate shift, $\text{MMD}^2(P_0, P_1) = 0 \iff P_0 = P_1$; with batch as “treatment”, the R -risk of the induced pseudo-outcome regression measures concept drift and is insensitive to pure covariate shift.

2 Path-order attribution and its non-uniqueness

Let $T(S)$ be a shift metric (MMD² or R -risk) evaluated on the feature subset $S \subseteq \{1, \dots, p\}$. Given an ordering π of the features, the *addition path* is the curve $S_0 = \emptyset \subset S_1 \subset \dots \subset S_p$ with $S_k = \{\pi_1, \dots, \pi_k\}$, and the marginal credited to feature π_k is $T(S_k) - T(S_{k-1})$. Sampling many random π and averaging the credits yields the Shapley value; but the credits along any single path are what a “how much did feature f contribute” question actually asks.

Proposition 1 (Total identifiable, path non-unique). *The endpoint $T(S_p) = T(\{1, \dots, p\})$ is invariant to the order π : the total shift is uniquely defined and estimable. The intermediate values $T(S_k)$, $0 < k < p$, and hence the per-feature marginals, are not invariant to π whenever the features are dependent; there is no order-free “contribution of feature f ”. Shapley restores uniqueness only by averaging over all orders, i.e. by adopting a convention, not by identifying a truth.*

Empirical demonstration. We draw random feature-addition paths on the repository’s banded-Toeplitz covariate-shift and concept-drift DGMs and record the metric at every step. Figure 1 shows the result: all paths converge to the same full-set total (red, order-invariant), but the accrual curves

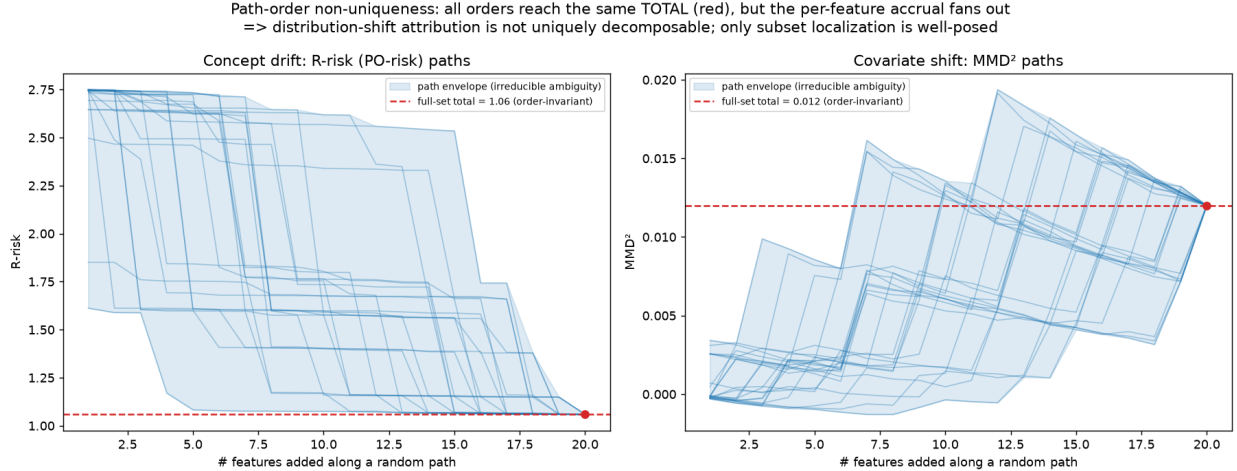


Figure 1: Random feature-addition paths. Every order reaches the same total (red, order-invariant); the per-feature accrual fans out (shaded envelope = irreducible ambiguity). Left: concept drift (R -risk); right: covariate shift (MMD^2).

fan out in between; the shaded band is the *irreducible attribution ambiguity*. For concept drift the maximum gap between two orders is $\approx$ the entire total (~ 1.0 against a total of ~ 1.06): two feature orderings can disagree about “how much has been explained so far” by essentially the whole shift. The per-step standard deviation across orders is 30–40% of the total for both metrics.

Remark 1 (Any number of levels). *The argument is scale-free: it applies to features within one entity, and equally to the coarser “how much does each child level contribute” question. Hence a distribution shift is not uniquely decomposable at any level of the hierarchy.*

3 The well-posed object: multi-level subset localization

While “how much” is not identifiable, “which subset carries the shift” is: the total is order-invariant, and the selection of the responsible group is stable across attribution conventions and across re-samples. We therefore localize recursively, applying the same two-stage selection at each level (a per-subset permutation test plus a bootstrap stability threshold, gated), and never report proportions:

1. **Detect** the shift globally (MMD / R -risk permutation test).
2. **Localize (level 1)** to a responsible attribute-group subset on the current axis.
3. **Drill (level 2)** by re-running the same selection on the next aggregation key (e.g. item, region), *conditioned* on the attribute flagged upstream — again returning a subset, not a share.
4. **Close the loop**: each selected subset carries an owner, a candidate rule (a KS-optimal split threshold), a decision (monitor / reweight / retrain / collect labels), and a trend status computed by diffing against the previous run.

4 A worked business example

Symptom. In the new week’s batch, the global test fires. Level-1 localization on the *region* axis selects the activity subset — concretely the order-count feature `n_orders` (and order-value aggregations) shifted *up* for a subset of users in region *R*: those buyers are placing more (and larger) orders than the existing batch.

Non-uniqueness, honestly. We do *not* claim “region *R*’s users account for 43% of the shift” — by the proposition that number is order/convention dependent. We claim the shift is *localized* to region *R*’s user subset, with a stability of, say, 0.98 and a split rule `n_orders per user > t`.

Drill one level. Re-running the same two-stage selection on the *item* axis, conditioned on the order-count signal, localizes the extra orders to a specific **product vertical** (item category *C*): within region *R*, the increase concentrates in category *C* (e.g. “fresh grocery”), not uniformly across the catalogue. Output: subset = {region *R*, user segment, category *C*} with its own split rule and stability.

Action / closed loop. The subset routes to the category-*C* / region-*R* owner; the candidate rule becomes a monitor; the decision (e.g. *reweight* if the mapping is unchanged, *collect labels* / *retrain* if the outcome relationship moved) is attached; the **trend** field marks it *resolved* once the subset stops flagging. The business learns *where* and *who* and *what to do* — never a spurious percentage.

Remark 2 (Why this is the right granularity). “43% from category *C*” would be *unfalsifiable* (non-identifiable) and un-actionable. “The new-batch order surge is localized to category *C* among region-*R* users; here is the threshold and the owner” is checkable, routable, and monitorable — it closes the loop.

5 How to evaluate such a system

Because per-feature proportions are not identifiable, they must not be the yard stick. Evaluate only identifiable targets:

Target	Identifiable?	Metric
Total / detection	yes (order-invariant)	test power; type-I under permuted labels
Subset localization	yes (convention-stable)	subset recall / precision vs. ground truth
Stability	yes	bootstrap selection frequency
Decision layer	yes (with labels)	downstream Δ loss backtest
Per-feature proportion	no	— (do not evaluate)

The principle: judge a method by the questions it claims to answer (detection, subset localization, stability, downstream impact), not by a question that is provably unanswerable. The non-identifiability itself is demonstrated empirically (Figure 1), not hidden.

6 Conclusion

The total distribution shift is identifiable; its per-feature decomposition is not. This is not a limitation of a particular estimator but a property of dependent features under any additive path

attribution. The productive response is to *localize responsible subsets* recursively down the entity hierarchy and attach owner / rule / decision / trend to each — turning an unanswerable “how much” into an operational “which subset, who, and what to do”.